

# SAURABH RAJ SHEKHAR

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## EDUCATION

**Netaji Subhas University of Technology (NSUT), Delhi**

*B.Tech, Computer Science & Data Science*

2024 – 2028

CGPA: 7.36

## TECHNICAL SKILLS

**Languages:** C++, Python, JavaScript, TypeScript, SQL

**Frameworks & Libraries:** React, Node.js, Express.js, FastAPI, Tailwind CSS

**AI / ML:** PyTorch, XGBoost, SHAP, ONNX Runtime Web, TensorFlow.js, Scikit-learn, GeoPandas, NumPy, Pandas

**Data & Infra:** DuckDB-WASM, Pyodide, MongoDB, Pinecone, Supabase, GeoParquet

**Tools & Practices:** Git, GitHub Actions (CI/CD), Vercel, Render, Vitest, Playwright, Postman

## PROJECTS

**AI Data Analyst Agent** | *React, TypeScript, DuckDB-WASM, Pyodide, Vercel Edge Functions*

[Live Demo](#)

- LLM writes SQL/Python against uploaded CSVs; validated and executed entirely client-side with a self-correcting retry loop (up to 3 attempts) — the model is never allowed to answer directly.
- Engineered a 5-provider LLM failover chain (Groq → Gemini → Mistral → Cerebras → Cohere) for zero-downtime fallback across free-tier rate limits; 208 passing tests with CI on every push.

**Agentic RAG Assistant** | *Node.js, React, TypeScript, Groq, Pinecone, Supabase*

[Live Demo](#)

- Built a tool-calling retrieval agent that decides for itself whether and how many times to search, backed by hybrid (vector + keyword) retrieval fused via Reciprocal Rank Fusion and LLM reranking.
- Added self-verification that checks generated answers against their cited sources and triggers a critique-guided follow-up search when unsupported; golden-set harness scores precision, faithfulness, and abstention.

**Vision Interpretability Studio** | *React, TypeScript, ONNX Runtime Web, TensorFlow.js, PyTorch*

[Live Demo](#)

- Trained a ResNet-18 from scratch (88.2% validation accuracy) and built 6 real-time interpretability views — exact zero-gradient Grad-CAM, Grad-CAM++, adversarial attacks, occlusion sensitivity — running entirely client-side.
- Hand-ported the model to TensorFlow.js for gradient-dependent techniques inference-only ONNX Runtime can't support, numerically verified to 6 decimal places against the PyTorch original; 57 tests (unit + Playwright e2e) in CI.

**HeatWatch India (Urban Heat Mitigation)** | *Python, FastAPI, XGBoost, SHAP, GeoPandas, React*

[Live Demo](#)

- Built a full-stack ML system analyzing urban heat islands across 20 Indian cities from Landsat satellite data; one XGBoost model with leave-cities-out cross-validation for honest generalization to unseen cities.
- Built SHAP-based driver analysis and an interactive Scenario Builder simulating cooling interventions with a live before/after map; submitted to ISRO Bharatiya Antariksh Hackathon (BAH) 2026.

## ADDITIONAL PROJECTS

- **Context Engineering Toolkit** — full context-engineering pipeline (selection, compression, summarization, evaluation) across 7 tools; 93 automated tests, CI. (*React, Node.js, Express*)
- **Cryptex** — token-based file-sharing platform with public/private visibility across 9 REST API modules. (*Node.js, Express, MongoDB, Supabase*)
- **Automata Lab** — zero-dependency DFA/NFA visualizer with live diagram rendering; used by NSUT CSDS peers for TAFL coursework. (*JavaScript, HTML, CSS*)

## CODING & OPEN SOURCE

- 235+ DSA problems solved in C++ across LeetCode, Code360, and GeeksforGeeks
- **claude-skills-vetted** — open-source catalog of 42 vetted Claude Skills with CI validation, install scripts, and issue templates

**Languages Known:** Hindi, English